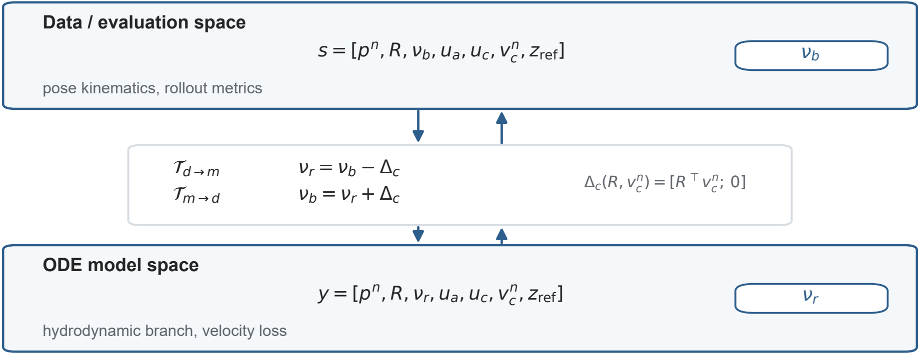
Data / evaluation space
$s = [p^n, R, \nu_b, u_a, u_c, v_c^n, z_{\mathrm{ref}}]$
ν_b
pose kinematics, rollout metrics
$\mathcal{T}_{d\to m}$
$\mathcal{T}_{m\to d}$
$\nu_r = \nu_b - \Delta_c$
$\nu_b = \nu_r + \Delta_c$
$\Delta_c(R, v_c^n) = [R^\top v_c^n;\ 0]$
ODE model space
$y = [p^n, R, \nu_r, u_a, u_c, v_c^n, z_{\mathrm{ref}}]$
ν_r
hydrodynamic branch, velocity loss